

ENGINEERING

# ServicePro Release Readiness

## 1. Release Classification

Publication edition | 2026-08-03

# Document Control

|                 |                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------|
| <b>Purpose</b>  | Enterprise platform overview and buyer evaluation reference                                 |
| <b>Audience</b> | Business leaders, platform administrators, evaluators, partners, and technical stakeholders |
| <b>Scope</b>    | Capabilities, architecture, security, deployment, adoption, outcomes, and terminology       |
| <b>Source</b>   | ServicePro repository documentation; technical meaning preserved                            |

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Beta → GA Requirements (All must pass)

Version: 8.0.0-alpha.1 Assessment Date: July 2026 Target: Alpha → Beta promotion gate

Current: Alpha (internal testing, single operator) Target: Beta (limited external customers, paid support) Future: GA (general availability, SLA-backed)

## 2. Gate Criteria Matrix

| #  | Criterion                        | Alpha (Current)    | Beta (Target)       | GA (Future)    |
|----|----------------------------------|--------------------|---------------------|----------------|
| 1  | Core features functional         | ■                  | Required            | Required       |
| 2  | Paid hosting (no cold starts)    | ■                  | Required            | Required       |
| 3  | Automated backups                | ■                  | Required            | Required       |
| 4  | CI test gating                   | ■                  | Required            | Required       |
| 5  | Error monitoring                 | ■                  | Required            | Required       |
| 6  | Uptime monitoring + alerting     | ■                  | Required            | Required       |
| 7  | Security audit (dependencies)    | ■                  | Required            | Required       |
| 8  | TypeScript strict mode           | ■                  | Recommended         | Required       |
| 9  | E2E test suite                   | ■                  | Required (5+)       | Required (50+) |
| 10 | Load testing                     | ■                  | Recommended         | Required       |
| 11 | Incident response runbook        | ■■■ Exists         | Required            | Required       |
| 12 | SLA definition                   | ■                  | Informational       | Required       |
| 13 | Data backup restoration verified | ■                  | Required            | Required       |
| 14 | Branch protection                | ■                  | Required            | Required       |
| 15 | HTTPS everywhere                 | ■                  | Required            | Required       |
| 16 | RBAC enforcement                 | ■                  | Required            | Required       |
| 17 | Multi-tenant isolation verified  | ■■■ Assumed        | Required (tested)   | Required       |
| 18 | API documentation                | ■■■ OpenAPI exists | Required (complete) | Required       |
| 19 | User documentation               | ■ 11 guides        | Required            | Required       |
| 20 | Payment processing               | ■                  | Required            | Required       |
| 21 | Email notifications              | ■                  | Required            | Required       |
| 22 | Customer portal functional       | ■■■ Partial        | Required            | Required       |

Current score: 5/22 required for Beta = Not ready for Beta promotion

## 3. Blocking Issues for Beta

### 3.1 Infrastructure (Must Fix)

| Issue                 | Current State         | Required State             | Effort              |
|-----------------------|-----------------------|----------------------------|---------------------|
| Cold start latency    | 50+ seconds           | <2 seconds                 | Upgrade Render plan |
| Database may pause    | 7-day inactivity rule | Always available           | Upgrade Supabase    |
| No backup restoration | Trust Supabase        | Verified restore procedure | 4 hours             |

| Issue                | Current State   | Required State     | Effort  |
|----------------------|-----------------|--------------------|---------|
| No uptime monitoring | Manual checking | Automated alerting | 2 hours |

### 3.2 Quality (Must Fix)

| Issue                  | Current State              | Required State                | Effort  |
|------------------------|----------------------------|-------------------------------|---------|
| Tests don't run in CI  | 770 tests exist, untested  | Green CI with coverage report | 4 hours |
| No type checking in CI | TypeScript errors possible | tsc --noEmit in CI            | 2 hours |
| No error tracking      | Errors invisible           | Sentry or equivalent          | 4 hours |
| No E2E tests           | Manual verification only   | 5+ critical path E2E tests    | 8 hours |

### 3.3 Security (Must Fix)

| Issue                      | Current State                 | Required State       | Effort  |
|----------------------------|-------------------------------|----------------------|---------|
| Dependency vulnerabilities | 3 high severity (web)         | Zero high severity   | 2 hours |
| No npm audit in CI         | Unknown vulnerability state   | Automated scanning   | 1 hour  |
| Static JWT secret          | Generated once, never rotated | Rotation mechanism   | 8 hours |
| No branch protection       | Direct push to main           | Required PR + review | 30 min  |

### 3.4 Feature (Must Fix for Beta)

| Issue                      | Current State            | Required State                 | Effort   |
|----------------------------|--------------------------|--------------------------------|----------|
| No payment processing      | Invoices without payment | Stripe integration             | 20 hours |
| No email notifications     | No transactional email   | SendGrid/Resend for key events | 12 hours |
| Customer portal incomplete | Login + basic views      | Full self-service              | 20 hours |

## 4. Verification Procedures

### 4.1 Pre-Release Checklist

- All CI checks passing (build, type check, tests, security scan)
- E2E smoke tests pass against staging
- Database backup taken and restoration verified
- No high/critical npm audit findings
- Error monitoring confirmed receiving events
- Uptime monitor confirmed alerting
- CORS configuration reviewed (production origins only)
- JWT secret is environment-specific (not shared across envs)
- Rate limiting tested under load
- Multi-tenant isolation manually verified (cross-tenant data leak test)
- OpenAPI spec matches actual endpoint behavior
- Customer-facing documentation reviewed for accuracy
- Rollback procedure documented and tested

### 4.2 Smoke Test Suite (Minimum for Beta)

| Test         | Endpoint     | Expected                 |
|--------------|--------------|--------------------------|
| Health check | GET /healthz | {"status":"ok"}          |
| Readiness    | GET /readyz  | 200 with DB connectivity |

| Test              | Endpoint                           | Expected                |
|-------------------|------------------------------------|-------------------------|
| Auth login        | POST /auth/login                   | JWT returned            |
| Auth refresh      | POST /auth/refresh                 | New JWT                 |
| Create customer   | POST /api/v1/customers             | 201 with customer ID    |
| List jobs         | GET /api/v1/jobs                   | 200 with array          |
| Public storefront | GET /api/public/storefront/aquapro | 200 with tenant profile |
| CORS preflight    | OPTIONS /api/v1/customers          | Correct headers         |
| Rate limit        | 100 requests in 1 second           | 429 after threshold     |
| Invalid auth      | GET /api/v1/customers (no token)   | 401                     |

### 4.3 Tenant Isolation Verification

1. Authenticate as tenant\_demo user
2. Attempt to access cd\_tenant\_demo customer data
3. Verify 403 or empty result (no cross-tenant leakage)
4. Repeat in reverse direction
5. Verify platform admin CAN access both (intended)

### 4.4 Database Recovery Test

1. Record current tenant count and recent customer
2. Trigger Supabase backup (or PITR snapshot)
3. Create new test data (customer, job)
4. Restore backup to fresh project
5. Verify pre-backup data intact, post-backup data absent
6. Document RTO (recovery time objective) achieved

## 5. Release Process

### 5.1 Current Process (Alpha)

Developer → Push to main → Render auto-deploy → Live

Problems: No review gate, no staging, no verification step.

### 5.2 Target Process (Beta)

Developer → Feature branch → PR (CI must pass) → Review → Merge to main  
 → Render staging deploy → Smoke tests (automated) → Manual approval  
 → Production deploy → Post-deploy verification

### 5.3 Rollback Procedure

| Step | Action                                                    | Time     |
|------|-----------------------------------------------------------|----------|
| 1    | Identify issue (Sentry alert or uptime monitor)           | 0–5 min  |
| 2    | Render dashboard → Manual Deploy → select previous commit | 2 min    |
| 3    | Verify rollback successful (health check green)           | 1 min    |
| 4    | If database migration was involved, run reverse migration | 5–15 min |
| 5    | Post-incident review                                      | 1 hour   |

Total expected rollback time: 3–8 minutes (no migration), 10–20 minutes (with migration)

## 6. Environment Matrix

| Environment    | Purpose                     | URL                             | Status         |
|----------------|-----------------------------|---------------------------------|----------------|
| Production API | Live traffic                | api.aardvark-enterprises.net    | Active         |
| Production Web | Live frontend               | app.aardvark-enterprises.net    | Active         |
| Staging        | Pre-production verification | —                               | Not configured |
| Local          | Developer machine           | localhost:3000 / localhost:3001 | Available      |

Recommendation: Configure staging on Render (same free tier or minimal paid) with separate Supabase project.

## 7. Monitoring & Alerting Plan

### 7.1 Uptime Monitoring

| Monitor       | URL                                | Check Interval | Alert Channel |
|---------------|------------------------------------|----------------|---------------|
| API Health    | GET /healthz                       | 5 min          | Email + SMS   |
| API Readiness | GET /readyz                        | 5 min          | Email         |
| Web Home      | GET /                              | 5 min          | Email         |
| Storefront    | GET /api/public/storefront/aquapro | 15 min         | Email         |

### 7.2 Error Alerting

| Trigger                     | Threshold         | Action                |
|-----------------------------|-------------------|-----------------------|
| Unhandled exception         | Any occurrence    | Sentry notification   |
| Error rate spike            | >10 errors/minute | Email alert           |
| Response time P95           | >5 seconds        | Warning               |
| Response time P95           | >15 seconds       | Critical alert        |
| Database connection failure | Any               | Critical alert + page |

### 7.3 Business Metrics (Post-Beta)

| Metric               | Source                     | Dashboard      |
|----------------------|----------------------------|----------------|
| Active tenants       | Database query             | Admin UI       |
| Jobs created/day     | Audit log                  | Dashboard      |
| Service requests/day | Public storefront          | Dashboard      |
| API requests/day     | Request metrics middleware | Render metrics |

## 8. Beta Launch Checklist

### Week -2 (Preparation)

- ☐ Upgrade Render to Starter plan
- ☐ Upgrade Supabase to Pro plan
- ☐ Configure Sentry project
- ☐ Set up UptimeRobot monitors
- ☐ Enable branch protection on main
- ☐ Run and fix npm audit
- ☐ Add TypeScript check to CI

## Week -1 (Verification)

- ☐ Run full test suite, document pass rate
- ☐ Execute E2E smoke tests
- ☐ Perform tenant isolation verification
- ☐ Verify database backup + restore
- ☐ Review CORS and security headers
- ☐ Test rollback procedure
- ☐ Update user documentation

## Launch Day

- ☐ Final smoke test pass
- ☐ Confirm all monitors green
- ☐ Verify error tracking receiving events
- ☐ Tag release v8.0.0-beta.1
- ☐ Communicate launch to early customers
- ☐ Monitor first 24 hours closely

## Week +1 (Stabilization)

- ☐ Review error reports from first week
- ☐ Address any P0/P1 issues discovered
- ☐ Collect early customer feedback
- ☐ Adjust rate limits if needed
- ☐ Verify billing/payments working (if enabled)

## 9. Known Risks at Launch

| Risk                            | Impact              | Mitigation                  | Owner |
|---------------------------------|---------------------|-----------------------------|-------|
| Supabase quota exceeded         | Service degradation | Monitor usage, upgrade plan | Ops   |
| Render instance memory pressure | OOM restart         | Monitor, optimize imports   | Dev   |

| Risk                            | Impact                 | Mitigation              | Owner    |
|---------------------------------|------------------------|-------------------------|----------|
| Phase stub modules slow startup | 3–5s added cold start  | Lazy-load or remove     | Dev      |
| Large component re-renders      | UI lag on slow devices | Component splitting     | Frontend |
| Single operator dependency      | Bus factor = 1         | Document all procedures | All      |

## 10. Post-Beta Roadmap Gates

### Beta → GA Requirements (All must pass)

- 1 30+ days uptime >99.5%
- 2 Zero data loss incidents
- 3 10+ active paying tenants
- 4 Load test: 100 concurrent users, P95 <2s
- 5 Security penetration test (third-party or automated)
- 6 SOC 2 Type I readiness assessment
- 7 50+ E2E tests passing
- 8 Full API documentation coverage
- 9 Customer support process defined
- 10 SLA published (99.9% uptime target)